Hugo Albert Bonet

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Education

University of Southern California (USC) — Los Angeles, CA
Master of Science in Computer Science, Current GPA: 3.92/4 — *Aug. 2025 – Expected May 2027*
- Fulbright Scholar (1 of 8 in Spain, sponsored by Iberdrola); USC Viterbi School of Engineering Scholarship.

Universitat Politècnica de València (UPV) — Valencia, Spain
Master of Science in Artificial Intelligence, GPA: 9.5/10 — *Sep. 2024 – July 2025*
Bachelor of Science in Data Science, GPA: 9.6/10 — *Sep. 2020 – June 2024*
- Honors: Best academic trajectory by NTT Data, national transcript award, best AI & mobility thesis by Renault.

Experience

Graduate Researcher (Robot Learning) — Dec. 2025 – Present
USC Learning and Interactive Robot Autonomy Lab (LiraLab) — *Los Angeles, CA*
- Designed an imitation learning algorithm in PyTorch and deployed it on a WidowX arm, reaching 83.6 IoU on real-robot Push-T vs. 73.4 for Implicit BC and 65.6 for Diffusion Policy; submitted to ICRA 2027.
- Reimplemented Diffusion Policy, Consistency Policy, and Implicit BC and ran 7-task sweeps on a SLURM GPU cluster; the method reached 93.9% on LIBERO-Goal vs. 84.5% for Diffusion Policy at 5.5× lower latency.
- Evaluated a fine-tuned $\pi_{0.5}$ vision-language-action (VLA) model on a real Franka Emika Panda arm across 4 manipulation tasks for a co-authored ICLR 2027 submission, whose method raised success from 20% to 60%.

Graduate Researcher (Computer Vision) — Nov. 2024 – Aug. 2025
Valencian Research Institute for Artificial Intelligence (VRAIN) — *Valencia, Spain*
- Co-developed a diffusion-based restoration web app for flood-damaged photographs, used by UPV's Conservation and Restoration department to restore 200,000 images with 98% less manual effort; published at AAAI-26.
- Built and labeled a 250-image training set, then fine-tuned SegFormer in PyTorch to replace a YOLO/SAM2 pipeline, cutting inference latency by 30% and raising expert-rated quality by 20%.

Research Intern (Reinforcement Learning) — Nov. 2023 – June 2024
VRAIN — *Valencia, Spain*
- Trained a Soft Actor-Critic levitation controller for Hyperloop UPV in a self-built Gymnasium simulator that transferred zero-shot to a test bench, at 0.06 ms on an STM32H7 with 94% less compute than cascaded PID.

Leadership

Team Lead — Aug. 2023 – Aug. 2024
Hyperloop UPV — *Valencia, Spain*
- Led a 48-member team that designed, built, and tested a 250 kg levitating vehicle, winning the Complete System Award at European Hyperloop Week 2024; wrote its C++ battery-management firmware.

Projects

Codenames Spymaster with SFT, DPO, and RL | *QLoRA, GRPO, Qwen2.5* — Aug. 2026 – Oct. 2026
- Fine-tuned Qwen2.5-1.5B with QLoRA on 2,000 synthetic examples, cutting invalid clues from 55.2% to 5.6% and raising correct guesses per turn from 0.31 to 1.20 on 1,000 paired boards; DPO and GRPO added no further gain.

LLM Watermarking Library | *PyTorch, vLLM, Hugging Face* — Aug. 2026 – Sep. 2026
- Designed an open-source KGW-style LLM watermarking library with adapters for Transformers, vLLM, llama.cpp, and SGLang, detecting 95% of 256-token passages at a 0.01% false-positive rate with 5.2% vLLM overhead.

Skills

Programming: Python, C++, SQL, JavaScript, HTML/CSS, MATLAB
ML Libraries: PyTorch, TensorFlow, Hugging Face, scikit-learn, pandas, MuJoCo, Gymnasium
Frameworks: vLLM, LangGraph, FastMCP, LlamaIndex, FastAPI, Flask
Developer Tools: Git, Linux, Docker, SLURM, pytest, Weights & Biases, AWS, Google Cloud, ONNX
Machine Learning: deep learning, computer vision, semantic segmentation, reinforcement learning, imitation learning
LLMs & Agents: SFT, DPO, GRPO, QLoRA, RAG, function calling, MCP, multi-agent workflows